

SalesOps automation: cost & benefit calculator

A live economic model for the SalesOps automation programme. The platform run cost is sized first from email volume, LLM token mix, OCR tier blend, and translation. That cost then converts directly into the three-year benefit case (FTE redeployment, annual net, ROI, payback, NPV) against the operational baseline below. Edit any input; every number recomputes instantly.

650 FTE in scope

\$15,000 fully-loaded cost / FTE

65% efficiency target

880k annual emails

3-year NPV horizon

o **Headline numbers**

Live, recompute on every input change

ANNUAL NET BENEFIT

\$3,587,797

Steady state, Year 2 onward

3-YEAR NPV (10% DISCOUNT)

\$5,696,902

Cumulative present value

ROI (3-YEAR)

185%

Return on total investment

PAYBACK

4.5 mo

Months to recoup implementation

Cost inputs · editable**VOLUME**

Annual inbound emails

Per Keysight Q&A baseline (~2k/day)

880000

Average attachments per email

Working assumption, refined at Functional Design

5

Average pages per attachment

Mix of PO PDFs, scans, embedded items

10

LLM TOKEN MIX (PER EMAIL)

Intake classification · input tokens

Email body + thread top, two-pass classifier

1100

Intake classification · output tokens

180

% of pages that get LLM extraction

PO-relevant pages only, not full attachment

35

Per-page extraction · input tokens

OCR text + per-intent schema prompt

1500

Per-page extraction · output tokens

Structured JSON of extracted fields

600

Decision call · input tokens

Context + four-gate scoring prompt

800

Decision call · output tokens

200

Reply drafting · input tokens

Resolved entities + reply template + glossary

1500

Reply drafting · output tokens

Customer-facing reply body

500

TRANSLATION

% of emails requiring translation

Non-English inbound + outbound reply

25

Avg characters per translation pass

Body + reply, both directions when needed

1800

UNIT PRICES (USD)

LLM family

Switches input, cached, and output unit prices

GPT-5.4



LLM input · \$ per 1M tokens

2.50

LLM cached input · \$ per 1M tokens

Used when system + glossary prompts hit cache

0.25

% of input tokens hitting prompt cache

50

LLM output · \$ per 1M tokens

15.00

OCR routing · % Read tier

\$1.50/1k pages · basic text on simple pages

80

OCR routing · % Layout tier

\$10.00/1k pages · PO line-item tables

18

OCR routing · % Custom tier

\$30.00/1k pages · trained Keysight templates

2

Translation · \$ per 1M characters

10.00

Reset cost defaults

Projected cost

ANNUAL TOTAL

\$348,453

\$0.3960 per email

MONTHLY

\$29,038

73,333 emails / month

PER EMAIL

\$0.3960

LLM + OCR + translate

COST BY COMPONENT (ANNUAL)

LLM input tokens

26.09B tokens / yr

\$35,877

10.3%

LLM output tokens

10.01B tokens / yr

\$150,216

43.1%

Azure DI OCR pages (blended)

44.0M pages / yr

\$158,400

45.5%

Translator (characters)	\$3,960	1.1%
396.0M chars / yr		

COST BY STAGE (ANNUAL)

Intake (classification)	\$3,707	1.1%
Extraction (OCR + LLM)	\$328,763	94.3%
Decision (four-gate)	\$3,608	1.0%
Communication (reply + translate)	\$12,375	3.6%

Where the spend lives: Extraction (OCR + LLM) is the dominant component at **94%** of annual spend (\$328,763). Per-email cost lands at **\$0.3960**. OCR routing is blended at **\$3.60 / 1k pages** (80% Read, 18% Layout, 2% Custom).

Benefit inputs · editable

OPERATIONAL BASELINE CURRENT

FTEs in scope	650
SalesOps order-management team, range 600 to 700	
Fully-loaded annual cost per FTE (USD)	15000
Salary, benefits, allocated overhead, tooling	
Current avg handle time per email (min)	12
Read, classify, look up, update, draft reply	
Productive hours per FTE per year	1700
2,080 raw minus PTO, training, meetings	

SOLUTION IMPACT TARGET

Operational efficiency target (%)	65
Share of current effort eliminated, range 60 to 70	
L4 auto-resolution rate (%)	35
Confidence above 0.95, no human touch	
L3 one-click review rate (%)	45
Operator approves draft action in one click	
L2 full HITL rate (%)	20
Operator handles edge case end-to-end	
L3 review time per email (min)	1
L2 HITL time per email (min)	5

INVESTMENT

Annual platform run cost (USD)	348453
Linked from Cost calculator; edit to override	
One-time implementation (USD)	1350000
Build, integrate, train, UAT, hypercare	
Annual support & governance (USD)	500000
L1/L2 support, evals, drift monitoring, KB curation	

VS COMPETITOR DELIVERY

ZBrain time-to-first-value (weeks) V1 cuts go-live to ~8 weeks via accelerated build	8
Competitor time-to-first-value (months) Industry baseline for enterprise SalesOps automation	6
Ramp-up after first value (months) Linear ramp from 0 to 100% of steady-state benefit	6
Comparison horizon (months)	36
Competitor one-time implementation (USD) Default matches ZBrain; edit if competitor pricing differs	1350000
Competitor annual support & governance (USD) Default matches ZBrain; edit if competitor pricing differs	500000
Competitor ramp duration (months) Linear 0 to cap; typical tier-1 SI runs 9 to 12 months to stabilize	9
Competitor steady-state cap (%) Realized benefit ceiling; default matches ZBrain. Drop below 100% to model a structural capability gap (less mature automation, slower iteration).	100

FINANCIAL ASSUMPTIONS

Year-1 benefit ramp (%) Pilot months 1 to 3, scale-out months 4 to 12	55
Year-2 benefit realization (%)	95
Year-3 benefit realization (%) Continuous-learning compounding	100
Discount rate (%) Cost of capital for NPV	10
Redeployment factor (%) Share of freed FTE that converts to cash; remainder redeployed	70

Reset benefit defaults

Print / Save as PDF

Summary

Year by year

Operational detail

Sensitivity

BASELINE COST (CURRENT)

\$9,750,000

TARGET FTE (POST-SOLUTION)

FTE FREED

423

650 FTE x \$15,000

227

At 65% efficiency

70% cash, 30% redeployed

ANNUAL RUN-RATE (STEADY STATE)

Gross FTE-equivalent value	\$6,337,500	value
Redeployed (non-cash)	-\$1,901,250	value
Cash savings	\$4,436,250	cash
Platform run cost	-\$348,453	expense
Support & governance	-\$500,000	expense
Annual net (steady state)	\$3,587,797	NET

TIERED AUTONOMY MIX



Headline: The solution converts **423 FTE** of operational effort (65% of the in-scope team) into **\$4,436,250** of annual cash savings after applying the 70% redeployment factor. Net of the \$348,453 platform run cost and \$500,000 annual support, that is **\$3,587,797 net per year** at steady state. Three-year NPV at 10.0% discount is **\$5,696,902**; implementation is recouped in **4.5 months**.

3 ZBrain vs competitor: time-to-value head start

Side-by-side cumulative cash position

CUMULATIVE HEAD START AT M36

+**\$1.80M**

Extra cash that ZBrain delivers

YEAR-1 NET CASH FLOW

\$2.41M

vs competitor \$655.2k

TIME-TO-VALUE LEAD

4.2 mo

Earlier first value

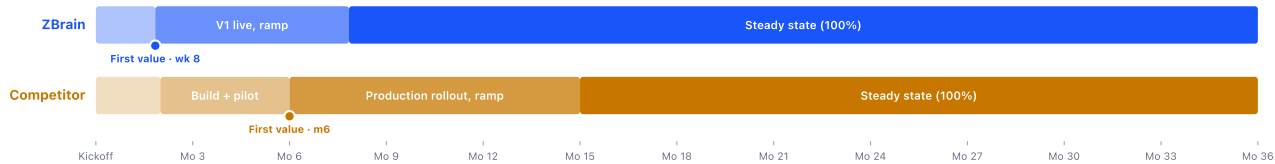
PAYBACK LEAD

6.0 mo

Earlier to break-even

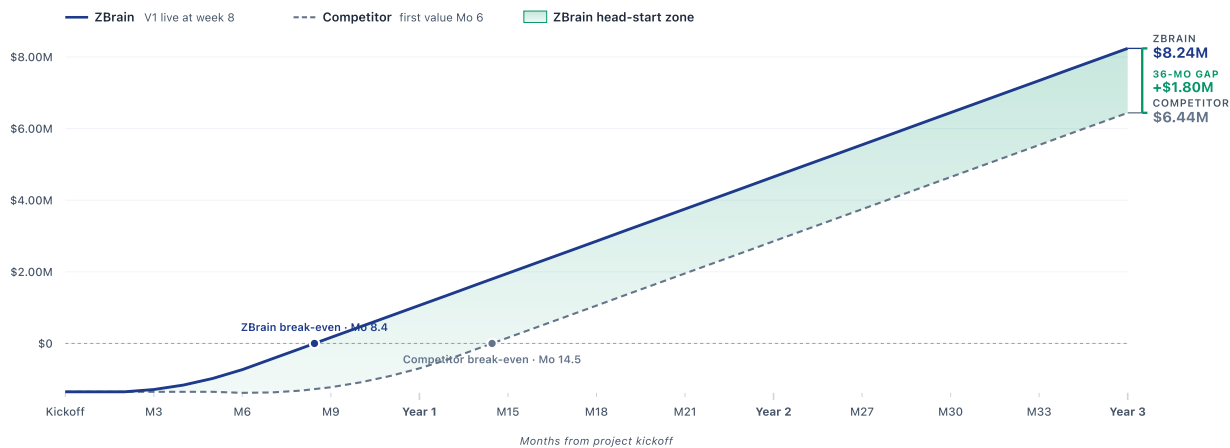
Project timeline · realized-benefit milestones

ZBrain's accelerated build puts V1 in production at week 8 and starts realizing benefit immediately. A tier-1 systems integrator follows a longer discovery / build / pilot cycle and does not begin realizing benefit until month 6, then runs a longer ramp.



Cumulative cash position over 36 months

Both projects pay the same implementation upfront. The shaded green area between the two lines is the cumulative head start ZBrain delivers month by month.



Assumptions used in this comparison

ZBrain V1 in production	Week 8 (accelerated build, pilot, UAT)
Competitor first value	Month 6 (typical tier-1 SI: discovery 2 mo, build + pilot 4 mo)
Ramp duration · ZBrain	6 months linear 0 to 100% of steady-state
Ramp duration · Competitor	9 months linear 0 to cap
Steady-state realization cap · ZBrain	100% (mature platform, continuous learning, rapid iteration)
Steady-state realization cap · Competitor	100% (matches ZBrain at the ceiling; the head-start gap is driven by slower ramp, not a lower cap)
Year-1 realized benefit · ZBrain	67% of steady-state run rate (8-week first value, 6-month ramp)
Year-1 realized benefit · Competitor	18% of steady-state run rate (month-6 first value, 9-month ramp, 100% cap)
Implementation cost · ZBrain	\$1,350,000 upfront
Implementation cost · Competitor	\$1,350,000 (matches ZBrain; edit to differ)
Annual run cost · ZBrain	\$348,453 platform + \$500,000 support & governance
Annual run cost · Competitor	\$348,453 platform + \$500,000 support & governance (matches ZBrain; edit to differ)
Annual cash benefit (full)	\$4,436,250 (FTE released x cost, x 70% redeployment)
Comparison horizon	36 months
Conservatism vs main case	Main Section 2 uses 55%/95%/100% Year-1/2/3 realization haircut on top. This view shows time-to-value directly.

Year-1 financial impact at 8-week start

The numbers below use the time-to-value model directly. The conservative 55% Year-1 realization shown in the main case (Section 2) is an additional risk haircut on top of these time-derived figures.

METRIC	ZBRAIN	COMPETITOR	DELTA
Time to first value	8 weeks	6 months	4.2 mo earlier

METRIC	ZBRAIN	COMPETITOR	DELTA
Payback (months from kickoff)	8.4	14.5	6.0 mo earlier
Year-1 net cash flow	\$2.41M	\$655.2k	+\$1.76M
Year-2 net cash flow	\$3.59M	\$3.55M	+\$41.1k
Year-3 net cash flow	\$3.59M	\$3.59M	+-\$0
Cumulative cash at m36	\$8.24M	\$6.44M	+\$1.80M

Year-1 reading: ZBrain delivers **\$2.41M** of net cash flow in Year 1 against the competitor's **\$655.2k**. The ZBrain Year-1 number is the time-derived 67% realization of the steady-state run rate, driven by the week 8 first value + 6-month ramp. The conservative 55% Year-1 figure used in Section 2 sits below this, providing a risk buffer. The 4.2-month time-to-value lead plus the slower competitor ramp compounds into a **\$1.80M** cumulative cash advantage by month 36.

Math · **Cost:** $\text{annual} = \text{LLM input} + \text{LLM output} + \text{Azure DI OCR} + \text{Translation}$; LLM input price is blended across cached and uncached at the cache-hit %. OCR is blended across Read / Layout / Custom by the routing percentages. **Benefit:** $\text{gross} = \text{FTE} \times \text{cost_per_FTE} \times \text{efficiency}$; $\text{cash} = \text{gross} \times \text{redeployment}$; $\text{net_year} = \text{cash} \times \text{ramp} - \text{platform} - \text{support}$; $\text{NPV} = \text{sum}(\text{net_year} / (1+r)^{\text{year}})$; $\text{payback} = \text{implementation} / \text{monthly_steady_net}$. Defaults reflect the proposal baseline.